

Test Procedure

The test procedure should include two files:

1. The test procedures (pdf)
2. An excel spreadsheet where the raw measurement data will be entered.

The test procedure should contain a section for each performance parameter to be characterized. Each section should include a list of equipment used to conduct the test and a description of each test to be conducted (with expected results), written for a test technician. You may assume that the technician knows how to use an oscilloscope (for basic signal capture), power supply, and signal generator but you should include setup details for each piece of equipment. Use the included test plan template as a starting point. For each test you should describe the setup, what data should be recorded, and how it should be analyzed. Your document should include all equations necessary to convert the raw data into the performance data.

You should not assume that the technician knows how to use the oscilloscope to measure Pmax-THD measurements - the setup and process for this measurement should be described in complete detail.

Your document should give test procedures for the parameters listed below. For all measurements, an 8 Ω load should be used and the volume should be set to the maximum position unless otherwise specified.

1. Maximum power output P_{MAX} (at which THD_F using the first 5 harmonics (V1-V5) at 1kHz is maintained below 1%). You should also record the corresponding input voltage (from signal generator) $V_{I_{MAX}}$, and the output voltage $V_{O_{MAX}}$.
2. Frequency response shown as gain in dB with input = $0.5 * V_{I_{MAX}}$. This should be plotted as gain in dB vs. log frequency from 20 Hz – 20 kHz. Phase should also be plotted. Include data for at least one point per octave (an octave is a doubling of the frequency).
3. Input and output impedance at 1 kHz. For this you may assume you have access to multiple load resistors from 4 to 16 ohms.
4. Amplifier efficiency defined as (output audio power)/(input power supply DC power) for input levels $V_{I_{MAX}}$ and $0.5 * V_{I_{MAX}}$ with the input frequency at 1 kHz.

Test Spreadsheet

The Excel spreadsheet should contain fields for entering all of the measurements described in the test procedure. It should also have formulas to calculate the performance parameters from the raw data. It should be formatted in a way that makes it clear to the person taking the data where each of the measurements should be entered and where the results will be displayed. There should be a field for each final calculation to be clearly displayed with correct units.